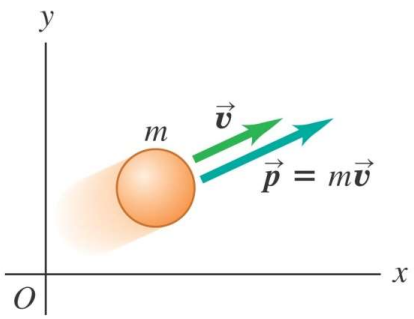
Linear momentum

Linear momentum

- In this lesson we will study linear momentum, which is useful for studying collision scenarios.
- Linear momentum is defined as $\vec{p} = m\vec{v}$

  where m is mass in kg and $\vec{v}$ is velocity in m/s. Hence the unit of linear momentum is kg m/s.
- The components of $\vec{p}$ are :

  $\vec{p}_x = m\vec{v}_x \quad \vec{p}_y = m\vec{v}_y \quad \vec{p}_z = m\vec{v}_z$



Momentum $\vec{p}$ is a vector quantity; a particle's momentum has the same direction as its velocity $\vec{v}$.

Newton's second law

- In terms of linear momentum, Newton's second law is

$$\Sigma\vec{F} = \frac{\mathrm{d}\vec{p}}{\mathrm{d}t} = \frac{\mathrm{d}(m\vec{v})}{\mathrm{d}t}$$

- In words, the net force acting on a particle equals the time rate of change of the momentum of the particle.
- A rapid change in momentum requires a large net force while a gradual change in momentum requires a small net force.
- If the mass m is constant and a is uniform, then $\Sigma\vec{F} = m\frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = m\vec{a}$

Impulse-momentum theorem

- If the net force is constant,

$$\Sigma \vec{F}\, \mathrm{d}t = \mathrm{d}\vec{p}$$

$$\int_{t_2}^{t_1} \Sigma \vec{F}\, dt = \int_{p_1}^{p_2} \mathrm{d}\vec{p}$$

$$\Sigma \vec{F}\,(t_2 - t_1) = (\vec{p}_2 - \vec{p}_1) = \vec{J}$$

- The term $\vec{J}$ is known as impulse. Its SI unit of impulse is N s.
- The above relation is known as the impulse-momentum theorem.

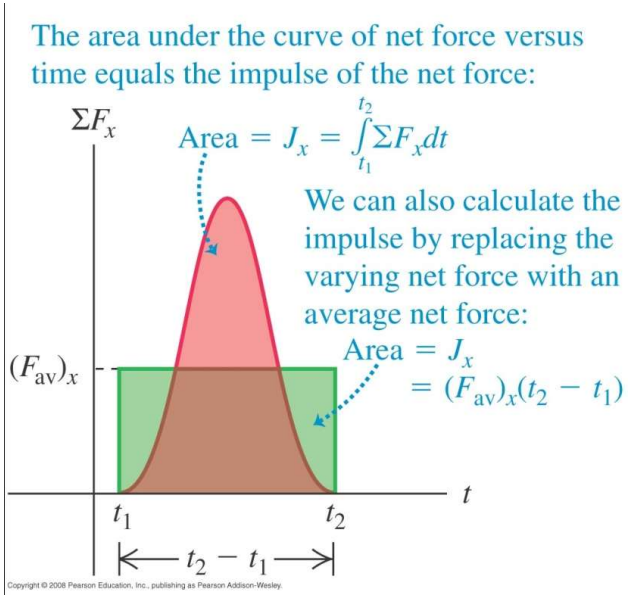
Impulse-momentum theorem

- If the net force is not constant, we can define an average net force which gives the same area as the varying net force, i.e.

$$\vec{J} = \vec{F}_{av}(t_2 - t_1)$$

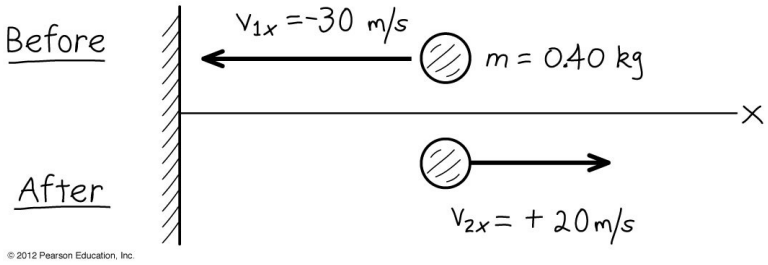
- The magnitude of the components of impulse are:

$$J_x = F_{av,x}(t_2 - t_1) = p_{2x} - p_{1x}$$
$$J_y = F_{av,y}(t_2 - t_1) = p_{2y} - p_{1y}$$
$$J_z = F_{av,z}(t_2 - t_1) = p_{2z} - p_{1z}$$



Example 1

You throw a ball with a mass 0.40 kg against a brick wall. It moves horizontally to the left at 30 m/s to hit the wall after which it rebounds horizontally to the right at 20 m/s. (a) Find the impulse of the net force on the ball during its collision with the wall. (b) If the ball is in contact with the wall for 0.010 s, find the average horizontal force that the wall exerts on the ball during the impact.



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(a) From the impulse-momentum theorem,

Impulse = change in momentum

$$= m(v_{2x} - v_{1x})$$

Taking right to be positive,

Impulse $= 0.4(20 - (-30)) = 20$ kg m/s to the right.

(b) Impulse $= F_{\text{av}}\Delta t$

$$F_{\text{av}}t = \frac{\text{Impulse}}{\Delta t} = \frac{20}{0.01} = 2000\,\text{N}$$

Internal and external forces

- The forces exerted by the particles within the system on each other are called internal forces.
- The forces exerted on the system of particles outside it are called external forces.
- If internal forces in a system are much larger than any external forces, we can treat the system as an isolated system.
- For example, when two pucks on an air table collide, the internal force between them is much greater than the frictional force between the air and the pucks.
- The two pucks together can be regarded as an isolated system.

Law of conservation of momentum

- The law of conservation of linear momentum can be shown using Newton's third law of motion, i.e. action equals reaction and acts in opposite directions.

- Consider an isolated system in which objects A and B collide.

$$\vec{F}_{A\ on\ B} = -\vec{F}_{B\ on\ A} \qquad \text{(Newton's third law)}$$

$$\vec{F}_{A\ on\ B} + \vec{F}_{B\ on\ A} = 0$$

$$\frac{d\vec{p}_B}{dt} + \frac{d\vec{p}_A}{dt} = 0 \qquad \text{(Newton's second law)}$$

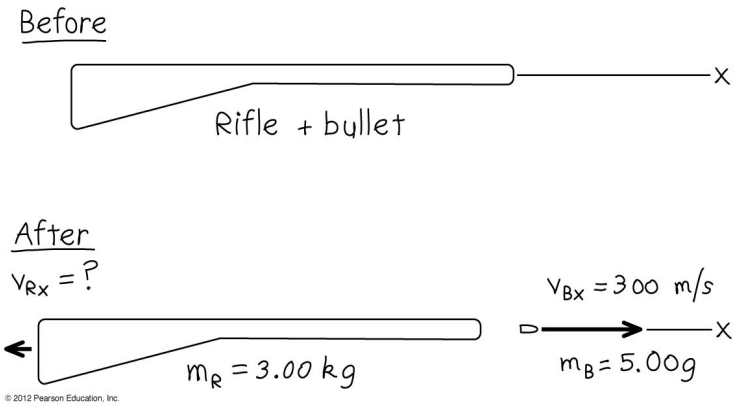
$$\frac{d(\vec{p}_A + \vec{p}_B)}{dt} = 0$$

$$\text{or } \vec{p} = \vec{p}_A + \vec{p}_B = \text{constant}$$

Hence, the total momentum of A and B is constant.

Example 2

A man holds a rifle of mass 3.00 kg loosely, so it can recoil freely. He fires a bullet of mass 5.00 g horizontally with a velocity 300 m/s (relative to the ground). (a) What is the recoil velocity of the rifle? (b) What are the final momentum and kinetic energy of the bullet and rifle?



(a) Since no external forces act on the rifle-bullet system we apply the law of conservation of momentum.

Before shooting, total momentum = 0 kg m/s

When shot is fired,

total momentum = $3.00 \times v_R + 0.00500 \times 300$

Hence, $0 = 3.00 \times v_R + 0.00500 \times 300$

$v_R = -0.500$ m/s (rifle recoils)

(b) Final momentum of rifle = $3.00 \times -0.500 = -1.50$ kg m/s

Final momentum of bullet = $0.00500 \times 300 = 1.50$ kg m/s

Final KE of rifle = $0.5 \times 3.00 \times 0.500^2 = 0.375$ J

Final KE of bullet = $0.5 \times 0.00500 \times 300^2 = 225$ J

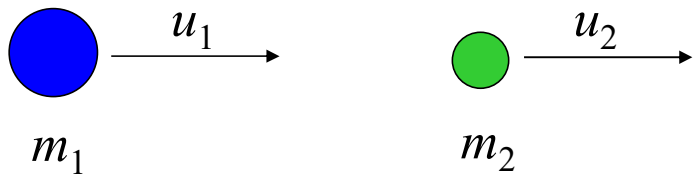
Elastic, inelastic and completely inelastic collision

- In an elastic collision, the total kinetic energy is conserved. The two objects separate after colliding.
- In an inelastic collision, the total kinetic energy is not conserved. The two objects separate after colliding.
- In a completely inelastic collision, the total kinetic energy is not conserved. The two objects stick together and move as one.

Note that total linear momentum is conserved whether the collision is elastic, inelastic or completely inelastic.

Elastic collision in one dimension

- Two spheres m_1 and m_2 moving in a straight line joining their centers collide elastically.



- Since the total linear momentum of the two spheres is conserved.

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$m_1(u_1 - v_1) = m_2(v_2 - u_2) \qquad \text{..... (1)}$$

Elastic collision in one dimension

- Since the collision is elastic, their total KE is also conserved.

$$\frac{1}{2}m_1u_1^2+\frac{1}{2}m_2u_2^2=\frac{1}{2}m_1v_1^2+\frac{1}{2}m_2v_2^2$$

$$m_1(u_1^2-v_1^2)=m_2(v_2^2-u_2^2) \qquad \text{..... (2)}$$

- Assuming $u_1 \neq v_1$ and $u_2 \neq v_2$

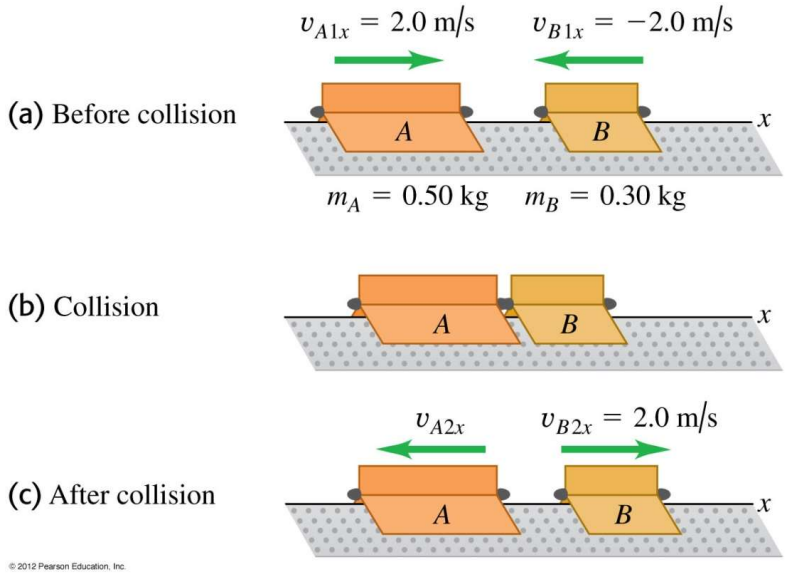
(2) ÷ (1) $\quad u_1+v_1=v_2+u_2$

or $\quad (u_1-u_2)=(v_2-v_1)$

- Hence for elastic collision, the relative velocity of approach before collision is equal to the relative velocity of separation after collision.

Example 3

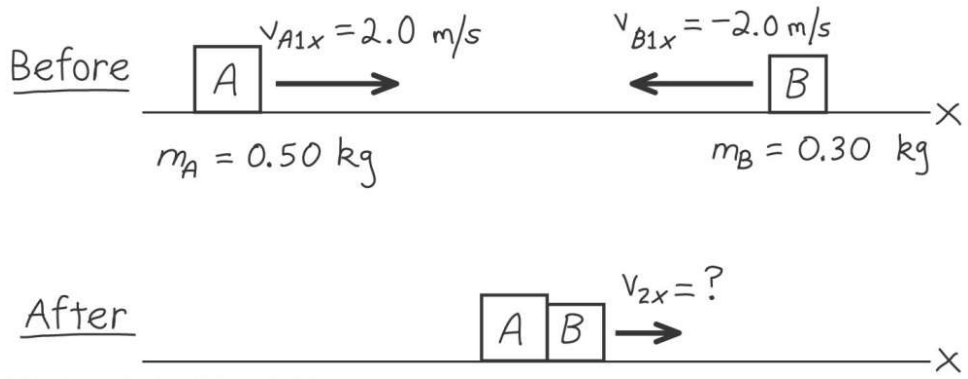
Two gliders with different masses move towards each other on a frictionless air track. After they collide glider B has a final velocity of +2.0 m/s (see figures). (a) What is the final velocity of glider A? (b) How do the changes in momentum and in velocity compare?



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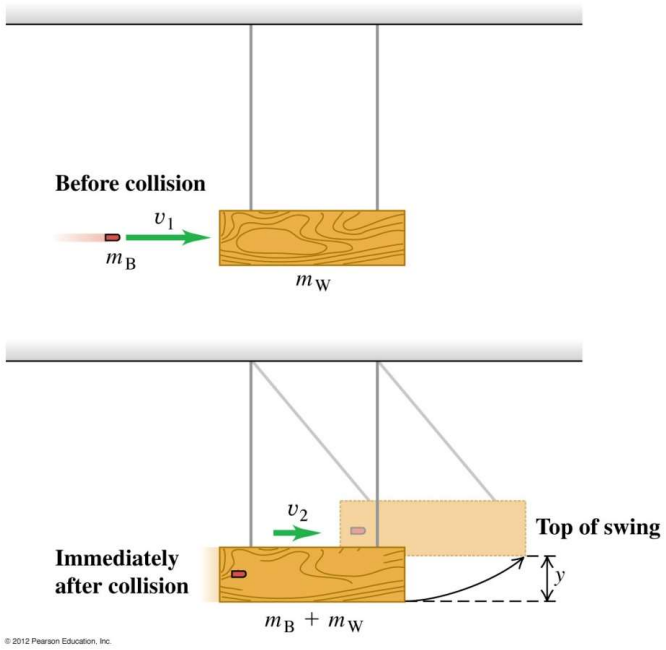
Example 4

Two gliders with different masses move towards each other on a frictionless air track. After they collide both of them stick together and move. Find (a) the common velocity and (b) compare their initial and final kinetic energies of the system.



Example 5

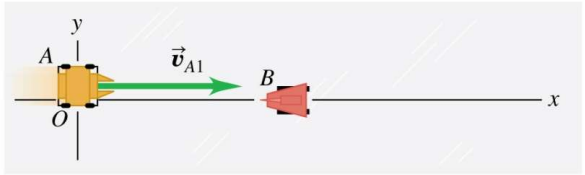
The below figure shows a ballistic pendulum, a simple system for measuring the speed of a bullet. A bullet of mass m_B makes a completely inelastic collision with a block of wood m_W which is suspended like a pendulum. After the impact the block swings up to a maximum height y. Find the initial speed of the bullet v_1 in terms of y, m_B and m_W and acceleration due to gravity g.



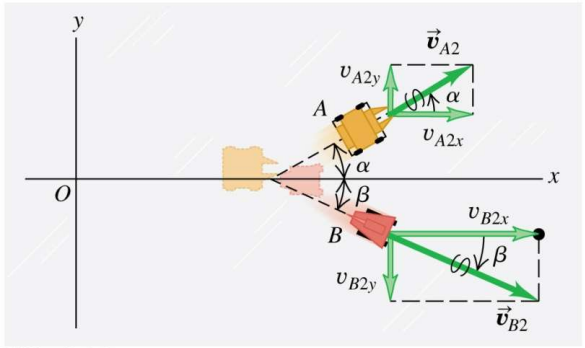
Example 6

The below figures show two robots namely A and B on a frictionless surface. Robot A with mass 20 kg initially moves at 2.0 m/s along x-axis towards robot B with mass 12 kg that is at rest. After the collision robot A moves at 1.0 m/s in a direction that makes an angle 30° with its initial direction. What is the final velocity of B?

(a) Before collision



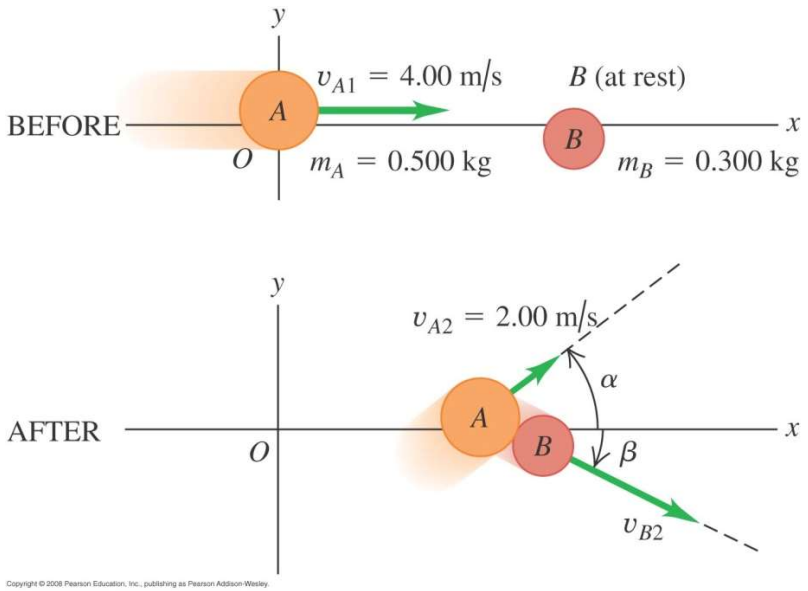
(b) After collision



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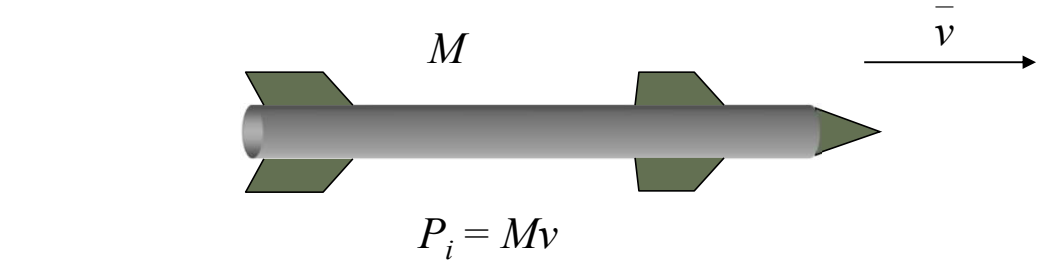
Example 7

The below figure shows an elastic collision of two pucks A and B of masses 0.500 kg and 0.300 kg respectively on a frictionless table. Puck A has an initial velocity of 4.00 m/s in the positive *x*-direction and a final velocity of 2.00 m/s in an unknown direction α. Puck B is initially at rest. (a) What is the final velocity of Puck B? (b) Find α and β.



Rocket propulsion

- A rocket flies forward by ejecting combustion gases backward.
- The mass of the rocket decreases with time as it burns up fuel.



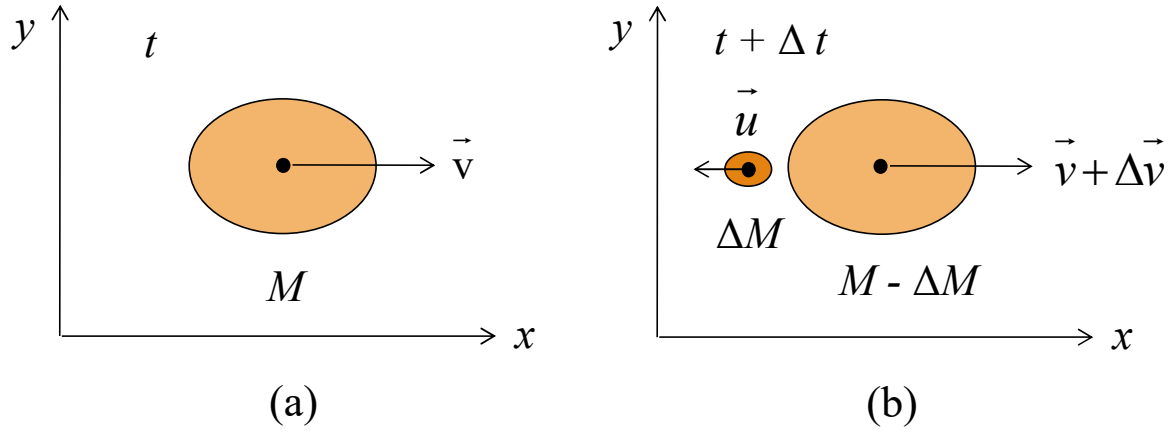
$$P_i = Mv$$



$$P_f = (M - \Delta M)(v + \Delta v)$$

Rocket propulsion

- Fig (a) and (b) below represent a rocket of mass M before and after exhaust gas of mass ΔM is ejected.
- The total momentum at time t is $M\vec{v}$
- The total momentum at time $t + \Delta t$ is $(M - \Delta M)(\vec{v} + \Delta v) + \Delta M\vec{u}$



Rocket propulsion

- Let the external force acting on the rocket be $\vec{F}_{\text{ext}}$.

$$\vec{F}_{\text{ext}} = \frac{[(M-\Delta M)(\vec{v}+\Delta\vec{v})+\Delta M\,\vec{u}]-M\vec{v}}{\Delta t} \qquad \text{from } \Delta\vec{p} = \vec{F}_{\text{ext}}\Delta t$$

$$= M\frac{\Delta\vec{v}}{\Delta t} + [\vec{u}-(\vec{v}+\Delta\vec{v})]\frac{\Delta M}{\Delta t}$$

As $\Delta t \to 0$,

i) $\dfrac{\Delta\vec{v}}{\Delta t} \to \dfrac{\mathrm{d}\vec{v}}{\mathrm{d}t}$

ii) $\dfrac{\Delta M}{\Delta t} \to -\dfrac{\mathrm{d}M}{\mathrm{d}t}$ for decreasing mass

iii) $\Delta\vec{v} \to 0$

Rocket propulsion

- Hence $\vec{F}_{\text{ext}} = M\dfrac{\mathrm{d}\vec{v}}{\mathrm{d}t} - [\vec{u} - \vec{v}]\dfrac{\mathrm{d}M}{\mathrm{d}t}$
- Rearranging, we get

$$M\frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \vec{F}_{\text{ext}} + [\vec{u} - \vec{v}]\frac{\mathrm{d}M}{\mathrm{d}t}$$

$$= \vec{F}_{\text{ext}} + \vec{v}_{\text{rel}}\frac{\mathrm{d}M}{\mathrm{d}t}$$

$$= \vec{F}_{\text{ext}} + \vec{F}_{\text{reaction}}$$

- Hence the thrust (reaction force) on the rocket depends on the exhaust speed relative to the rocket and the rate of change of mass (burn rate).

Example 7

A rocket stands vertically on its launching pad prior to ignition. The mass of the rocket and its fuel is 1.9 x 10^3 kg. On ignition, gas is ejected from the rocket at a speed of 2.5 x 10^3 kg/s relative to the rocket and fuel is consumed at a constant rate of 7.4 kg/s. Find the thrust on the rocket and hence explain why there is an interval between ignition and lift-off.

$$M\frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \vec{v}_{rel}\frac{\mathrm{d}M}{\mathrm{d}t}$$

$$= 2.5\times10^{3}\times7.4$$

$$= 1.9\times10^{4}\,\mathrm{N}$$

The delay is due to the inertia of the rocket and initial fuel load.

End of chapter